US HOUSE PRICE ANALYSIS

PROJECT REPORT II-ADVANCED ANALYSIS

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Abstract

The real estate market is a cornerstone of the U.S. economy. Housing prices are particularly important as they reflect broader economic conditions and impact financial markets. Understanding these factors is essential for buyers, sellers, and policymakers to make informed decisions. This report focuses on the exploration to understand the dynamics of housing prices, with a specific focus on creating a predictive model that not only delivers accurate forecasts but also offers clear interpretations of the underlying dynamics. Through this study, location emerges as the primary factor influencing house prices, particularly in Washington, D.C., highlighting the importance of strategic location planning when purchasing or investing in real estate. Additionally, this study introduces a model-driven recommendation system that identifies the best housing options within a specified price category, enhancing decision-making for prospective buyers.

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Introduction

In today's world, real estate is a worthy investment in any part of the world. The U.S. real estate market, in particular, holds a renowned place due to the country's immense wealth, its effect on the world economy and financial stability. Identifying the primary factors that drive housing prices is essential for various stakeholders, including investors, buyers, and real estate developers. By understanding the dynamics of housing prices, investors can identify profitable opportunities, and buyers can make informed purchasing decisions. This report embarks on a journey to unravel the intricate dynamics of housing prices, with a focus on constructing a predictive and interpretable model. Designed to offer a comprehensive insight into the factors that influence house prices, and how location affects the built-in properties of the houses, this study serves as a valuable resource for buyers and investors alike.

Description of the Question

The data set has been analyzed to explore the following three primary questions:

1. What are the key factors that influence house prices within this data set?
2. How do house prices compare across low, mid, and high ranges, and what features contribute to achieving the best price within each category?
3. In what ways has the location affected the built-in properties of the houses? Are there any significant patterns that emerge from this relationship?

Description of the Dataset

This study is based on housing market data from 2014, offering insights into historical trends and potential factors influencing property prices. The data set sourced from Kaggle pertains to house prices in the United States, collected during the year 2014. It includes 18 distinct variables and was compiled between May 2 and July 7, during which a total of 4,600 house prices were recorded.

Variable	Definition	Variable	Definition
Date	The date the house was sold.	View	Quality of the view (0 = No special view, 1 = Slightly better, 2 = Moderate, 3 = Good view, 4 = Excellent view).
Price	The price of the house.	Condition	Condition of the house (1 to 5 scale, higher is better).
Bedrooms	Number of bedrooms.	Sqft_above	Square footage of the house above ground.
Bathrooms	Number of bathrooms (1 = full bathroom, 0.5 = half bathroom, 0.25 = quarter bathroom).	Sqft_basement	Square footage of the basement.
Sqft_living	Living area in square feet (sqft_above + sqft_basement).	Yr_built	Year the house was built.
Sqft_lot	Lot size of land in square feet.	Yr_renovated	Year the house was last renovated (0 if never renovated).
Floors	Number of floors (1 = full floor, 0.5 = small upper area).	Street	Street address.
Waterfront	Indicator (0 = No, 1 = Yes) for waterfront property.	City	City location.
		Statezip	State and ZIP code.
		Country	Country (all entries seem to be 'USA').

Table 1: Description of Variables

Important Results of the Descriptive Analysis

Cluster analysis and factor analysis were performed as a part of the advanced analysis as a completion of the Descriptive Analysis. The findings include the following,

• There were no significant clusters formed among the observations, as backed up by the silhouette score. The optimal number of clusters was determined to be 2; however, this result did not effectively form distinct clusters, yielding values of only 0.4 and 0.26. Thus, the analysis using clusters was deemed ineffective.

• PCA and FAMD suggested that House_Age and Sqft_lot have no significant contribution to the predictor space.

• The formation of clusters based on factor scores was not considered as viable as it did not provide any meaningful interpretation.

• Most houses in the area do not have a basement due to high sea water levels; however, this was contradicted by houses in Seattle, where the numbers of houses with and without basements were roughly equal.

• The City and Statezip variables were found to be highly correlated.

Important Results of the Advanced Analysis

To explore the key factors influencing house prices within this dataset, several models were fitted, including Ridge, Lasso, Elastic Net, Generalized Boosting Method (GBM), XG Boost, and Random Forest. The corresponding results are presented below.

Model	Training MSE	Training R^2	Test MSE	Test R^2
Ridge	0.2468	0.7532	0.3127	0.6869
Lasso	0.1668	0.8332	0.2349	0.7648
Elastic Net	**0.1667**	**0.8333**	**0.2346**	**0.7650**
GBM	0.1866	0.8134	0.2761	0.7234
XGBoost	0.0610	0.9389	0.2597	0.7399
Random Forest	0.0558	0.9441	0.3298	0.6697

Table 2: Summary of All R^2 & MSE Values

Optimal model: Elastic net

Elastic Net has the lowest Test MSE (0.2346) and the highest Test R^2 (0.7650), making it the best-performing model in terms of generalization. It also has a Training MSE of 0.1667 and a Training R^2 of 0.8333, indicating that it fits the training data well while maintaining strong performance on the test set. This suggests that Elastic Net strikes a good balance between bias and variance, avoiding both overfitting (where the model performs well on training data but poorly on test data) and underfitting (where the model fails to capture patterns in the data).

Compared to Elastic Net, Ridge regression has a higher Test MSE (0.3127) and lower Test R^2 (0.6869), indicating weaker generalization due to its inability to perform feature selection. Lasso performs similarly to Elastic Net but may over-penalize some coefficients, making Elastic Net more stable. XGBoost (Test MSE = 0.2597, Test R^2 = 0.7399) and GBM (Test MSE = 0.2761, Test R^2 = 0.7234) show strong predictive power but exhibit higher variance, suggesting potential overfitting. Random Forest has the highest Test MSE (0.3298) and lowest Test R^2 (0.6697), indicating severe overfitting.

Feature Importance Analysis

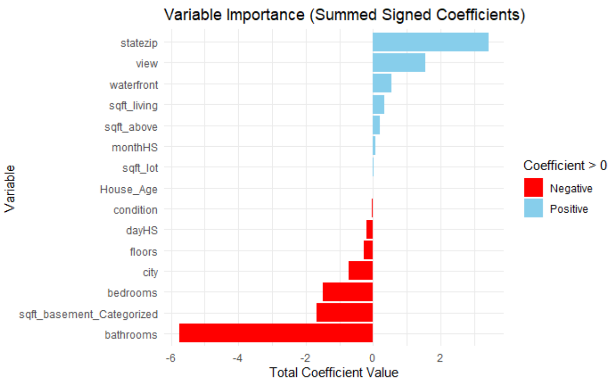

Figure 1: Variable Importance Plot

#	Feature	Value
1	Bathrooms	-5.740095153
13	Statezip	3.420542462
10	Sqft_basement_Categorized	-1.676516626
14	View	1.561203511
2	Bedrooms	-1.485022692
3	City	-0.731666230
15	Waterfront	0.553542616
11	Sqft_living	0.336066420
6	Floors	-0.264668143
9	Sqft_above	0.214863290

Table 3: Values of Sum of Coefficients

Top 10 Features

Feature	Value
bathrooms.6.25	-2.1580762
bathrooms.8	-2.1303919
sqft_basement_Categorized.9	-1.0820940
statezip.WA 98112	0.9701001
statezip.WA 98109	0.9562965
city.Medina	0.9502969
statezip.WA 98178	-0.8801969
statezip.WA 98119	0.8033137
statezip.WA 98004	0.7895679
statezip.WA 98105	0.7524778

Table 4: Top 10 Features

Price Increasing Factors: The most significant factor influencing house prices is location**,** with high-value ZIP codes like WA 98112 and WA 98109 and cities like Medina driving prices up. Homes with luxury features such as waterfront views and scenic surroundings also see a notable increase in value. Additionally, larger living spaces**,** particularly in terms of total square footage and above-ground area, positively impact home prices**.**

Price Decreasing Factors: Having too many bathrooms, especially 6 or more, tends to decrease home values, likely due to inefficiency in space usage. Similarly, large basements appear to be less desirable, possibly because they are often unfinished or not as functional. More bedrooms and extra floors can also have a negative impact in some cases, suggesting that buyers prioritize open and well-utilized space over sheer room count.

House age does not appear to be a significant factor in determining home prices

Classification

In order to accomplish the 2nd objective of the study, Classification was performed using a Random Forest model, in which the response variable was modified as house price categories. In the descriptive analysis, it was suggested to use four categories: LOW, MID, HIGH, and LUXURY. However, due to a significant under-sampling of observations in the LOW category, this classification was misclassified at a rate of 100%. As a result, the categorization was adjusted to three categories: MID, HIGH, and LUXURY.

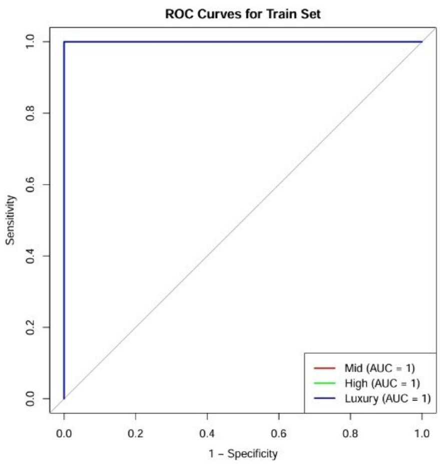

Figure 3: ROC Curve for Train Set

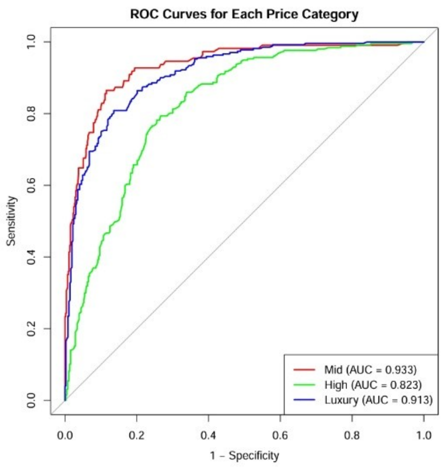

Figure 2: ROC Curve for Test Set

The ROC curves were generated following hyperparameter tuning. Although the model was overfitted to the training set at 100%, it still classified the test set with an accuracy close to 90% across all levels, thus making it unignorable. No improvements were observed for the less observed category, even with the application of weightings.

Following the classification, a system of models was developed to recommend the best house within a specified price category. The houses were filtered based on user-defined criteria, and a hybrid scoring approach was implemented, combining a custom scoring system with a machine learning scoring system to determine the best house.

bedrooms <fctr>	bathrooms <fctr>	sqft_living <dbl>	sqft_lot <dbl>	floors <fctr>	waterfront <fctr>	view <fctr>	condition <fctr>	sqft_above <dbl>	sqft_basement <dbl>	city <fctr>	House_Age <dbl>	monthHS <fctr>	dayHS <fctr>	sqft_basement_Categorized <fctr>
5	3.5	5960	13703	2	0	2	3	4770	1190	Bellevue	30	05	29	3

1 row | 1-15 of 17 columns

sqft_living <dbl>	sqft_lot <dbl>	floors <fctr>	waterfront <fctr>	view <fctr>	condition <fctr>	sqft_above <dbl>	sqft_basement <dbl>	city <fctr>	House_Age <dbl>	monthHS <fctr>	dayHS <fctr>	sqft_basement_Categorized <fctr>	price_category <fctr>	score <dbl>
5960	13703	2	0	2	3	4770	1190	Bellevue	30	05	29	3	High	1790.1

1 row | 3-17 of 17 columns

Figure 4: Example Usage of the Recommendation System

Spatial Analysis

A thorough spatial analysis using latitude and longitude suggests that the previously found observations related to IOWA and San Antonio were misinterpreted and that all observations originated from the state of Washington.

The Random Forest analysis suggests a significant impact on the variable city. Thus, in order to meet the 3rd objective, a geographically weighted regression was performed and tested using only the numerical data.

```
Summary of GWR coefficient estimates at data points:
                   Min.     1st Qu.      Median     3rd Qu.        Max.      Global
X.Intercept. -9.8101e+07 -8.5921e+07 -7.9441e+07 -6.9061e+07 -2.9611e+07 -5.1442e+07
sqft_living   2.0627e+02  2.1614e+02  2.2025e+02  2.2961e+02  3.3318e+02  2.5687e+02
sqft_lot     -7.4140e+00 -7.1576e+00 -6.8728e+00 -6.4281e+00 -3.5721e+00 -5.9839e+00
sqft_above    1.0077e+01  2.3086e+02  2.5224e+02  2.6387e+02  2.8289e+02  1.8706e+02
House_Age     3.7539e+03  4.4798e+03  4.5886e+03  4.6864e+03  4.7844e+03  4.6960e+03
lat           5.3876e+05  7.1071e+05  7.2534e+05  7.5412e+05  7.9947e+05  7.1981e+05
lng          -4.9320e+05 -4.1798e+05 -3.6322e+05 -2.8399e+05 -1.4061e+03 -1.3800e+05
Number of data points: 77
Effective number of parameters (residual: 2traceS - traceS'S): 11.17765
Effective degrees of freedom (residual: 2traceS - traceS'S): 65.82235
Sigma (residual: 2traceS - traceS'S): 255688.8
Effective number of parameters (model: traceS): 9.587719
Effective degrees of freedom (model: traceS): 67.41228
Sigma (model: traceS): 252655.6
Sigma (ML): 236403
AICc (GWR p. 61, eq 2.33; p. 96, eq. 4.21): 2148.93
AIC (GWR p. 96, eq. 4.22): 2133.591
Residual sum of squares: 4.303251e+12
Quasi-global R2: 0.7659734
```

Figure 5: Summary Result of Geographically Weighted Regression Model

The intercept median is -7.9441e+07. This suggests that the base price of a house, without considering predictors, varies significantly across different locations. This can be further backed by the results from random forest fittings, which indicate a substantial impact from the city variable.

Issues Encountered and Proposed Solutions

During our advanced analysis, several challenges were encountered. At the initial stage, inconsistencies in the data set, such as missing values, posed a challenge. To address this issue, Random Forest (RF) imputation was utilized to handle the missing values. Additionally, 2 outliers were identified using the isolation forest method; however, these were removed as they were determined to be data entry errors.

Furthermore, the system of models developed to recommend the best house within a specified price category based on predefined score ratings needs improvement. The model's performance was negatively impacted by the structural issues arising from using three consecutive models to classify, score, weight, and predict based on user input, leading to a lack of coherence in finding the best house. To enhance model performance in identifying the best houses, we recommend employing a trained neural network, which is expected to yield a higher prediction rate. Additionally, the model should be improved to provide customers with options that align with their preferences.

Discussion

After completing the EDA, the focus was redirected to addressing data inconsistencies in a more comprehensive manner. Missing values were handled using Random Forest (RF) imputation, and two outliers, identified through the isolated forest method, were removed as they were determined to be data entry errors. Furthermore, the date variable was replaced by two new variables: 'DayHS' and 'MonthHS' to facilitate further analysis.

A cluster analysis was performed using K-Means clustering, Hierarchical clustering and DBSCAN algorithm. However, the formation of clusters was not considered as viable as it did not provide any meaningful interpretation.

The initial steps of the advanced analysis involved implementing Ridge regression, Lasso regression, and Elastic Net regression. Multiple Linear Regression, although a straightforward model, was not implemented due to the presence of multicollinearity in the data.. Subsequently, tree-based methods such as the Generalized Boosting Method (GBM), XG Boost, and Random Forest were implemented to determine the optimal model for predicting and interpreting the data. Ultimately, Elastic Net regression was identified as the optimal model.

A feature importance analysis was performed to determine which variables have the most influence on house prices. The analysis found that location is the most significant factor positively influencing home values, while having an excessive number of bathrooms tends to decrease those values. Overall, house prices are primarily affected by location, desirable features, and functional space, while overbuilt or impractical elements can negatively impact a property's value.

Additionally, the potential for classifying houses by city was explored using a random forest model. The original price variable was categorized into the following price ranges:

- Mid: $0 – $299,999

- High: $300,000 – $499,999

- Luxury: $500,000 – $1,000,000,000

However, due to the large number of categories (77), the model observed a significant reduction in performance, approximately 0.44%. The sensitivity for most cities was near zero, leading to the decision to deprecate the model.

The system designed to recommend the best house within a specified price range uses a hybrid scoring approach. This approach combines scores from machine learning models, specifically prediction probabilities from Random Forest and XGBoost. Additionally, a custom scoring system is implemented, which assigns weighted scores based on different price categories: Mid, High, and Luxury. Finally, machine learning predictions were merged with the custom scores by calculating the overall score as follows: (ML Prediction Score + Weighted Custom Score) / 2. The house with the highest final score is then recommended to the user.

The spatial regression was not performed using categorical data due to the fact that it relies on continuous predictors. Instead, an approach like one-hot encoding should be utilized for mixed data. However, due to the computational complexity and sparsity issues associated with a large number of categories, this was not implemented. Further analysis is recommended using a large data set that includes more geospatially influenced features, such as the radius of houses around the city, etc. The median coefficient of House_Age suggests that the price is influenced by increases in house age, which was not considered due to a lack of evidence.

Conclusion

The real estate market is a complex puzzle, but this study pieces together the key factors that shape housing prices. At the heart of it all? Location. Whether it's a waterfront, view, or a coveted ZIP code, where a house stands has the most significant impact on its value. Through comprehensive analysis, Elastic Net emerged as the most reliable model for predicting prices, balancing precision with interpretability. Feature analysis revealed surprising insights. While luxury features drive prices up, too many bathrooms and excessive bedrooms can actually decrease a home's value. Beyond prediction, this study developed a smart recommendation system, helping buyers find the best homes in their price range. Ultimately, this research confirms that real estate is more than just numbers; it involves finding the perfect balance between location, features, and smart investment.

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Appendix

R Code – Elastic Net Model Generation

```r
#Please change the Directory as needed.
library(glmnet)
library(caret)
library(readr)
train <- read_csv("train_RFIMPUTED.csv")
test <- read_csv("test_RFIMPUTED.csv")
# Factorize categorical variables
categorical_vars <- c("bedrooms", "bathrooms", "floors", "waterfront",
                      "view", "condition", "city", "monthHS", "dayHS",
                      "sqft_basement_Categorized", "statezip")
train[categorical_vars] <- lapply(train[categorical_vars], as.factor)
test[categorical_vars] <- lapply(test[categorical_vars], as.factor)
continuous_vars <- c("sqft_living",
  "sqft_lot", "price", "sqft_above", "sqft_basement")
continuous_vars <- c("sqft_living","price",
  "sqft_lot","sqft_above", "sqft_basement","House_Age")
# Scale the numerical variables
train[continuous_vars] <- scale(train[continuous_vars])
# Scale the numerical variables
test[continuous_vars] <- scale(test[continuous_vars])
#ElasticNet
library(glmnet)
library(caret)
# Specify continuous and categorical variables
continuous_vars <- c("sqft_living", "sqft_lot",
    "price", "sqft_above", "sqft_basement", "House_Age")
categorical_vars <- c("bedrooms", "bathrooms", "floors", "waterfront",
                      "view", "condition", "monthHS", "dayHS", "city",
                      "sqft_basement_Categorized", "statezip")# Exclude 'statezip'
# One-hot encode categorical variables (excluding 'statezip')
dummies <- dummyVars(price ~ ., data = train[,
      c(categorical_vars, continuous_vars)], fullRank = TRUE)
x_categorical <- predict(dummies, newdata = train)
# Combine continuous and encoded categorical variables
x_continuous <- as.matrix(train[,
        continuous_vars[continuous_vars != "price"]])# Exclude price from predictors
x <- cbind(x_continuous, x_categorical)  # Final predictor matrix
x <- x[, !duplicated(colnames(x))]
# Response variable
y <- train$price
# Fit Elastic Net Regression
set.seed(1234)
cv.elastic <- cv.glmnet(x, y, alpha = 0.5)
# Alpha = 0.5 for Elastic Net (mix of Ridge and Lasso)
plot(cv.elastic)
bestlam_elastic <- cv.elastic$lambda.min
cat("Best lambda for Elastic Net:", bestlam_elastic, "\n")
# Fit Elastic Net model with the best lambda and view coefficients
final_elastic <- glmnet(x, y, alpha = 0.5, lambda = bestlam_elastic)
coef(final_elastic)
# Predict on the test set using the final Elastic Net model
x_test_categorical <- predict(dummies, newdata = test[, c(categorical_vars,
                                                        continuous_vars)])
x_test_continuous <- as.matrix(test[, continuous_vars[continuous_vars != "price"]])
x_test <- cbind(x_test_continuous, x_test_categorical)
x_test <- x_test[, !duplicated(colnames(x_test))]
y_pred_elastic <- predict(final_elastic, s = bestlam_elastic, newx = x_test)
# Compare the predictions to the actual values
y_actual <- test$price
# Calculate Mean Squared Error (MSE)
mse_elastic <- mean((y_pred_elastic - y_actual)^2)
cat("Test (MSE) for Elastic Net:", mse_elastic, "\n")
# Calculate R-squared for Test Data
ss_total_test <- sum((y_actual - mean(y_actual))^2)
ss_residual_test <- sum((y_actual - y_pred_elastic)^2)
r2_test <- 1 - (ss_residual_test / ss_total_test)
cat("Test R^2 for Elastic Net:", r2_test, "\n")
# TRAINING MSE for Elastic Net
y_train_pred_elastic <- predict(final_elastic, s = bestlam_elastic, newx = x)
mse_train_elastic <- mean((y - y_train_pred_elastic)^2)
cat("Training MSE for Elastic Net:", mse_train_elastic, "\n")
# Calculate R-squared for Training Data
ss_total_train <- sum((y - mean(y))^2)
ss_residual_train <- sum((y - y_train_pred_elastic)^2)
r2_train <- 1 - (ss_residual_train / ss_total_train)
cat("Training R^2 for Elastic Net:", r2_train, "\n")
# Plot residuals for the training set
residuals_train <- y - y_train_pred_elastic
plot(residuals_train, main = "Residuals for Training Data",
     ylab = "Residuals", xlab = "Index")
abline(h = 0, col = "red")
###############################
# Get coefficients for the final Elastic Net model
coefficients <- coef(final_elastic, s = bestlam_elastic)
# Convert the coefficients to a data frame for easier manipulation
coefficients_df <- data.frame(Feature = rownames(coefficients),
                              Coefficient = as.vector(coefficients))
# Remove the intercept row
coefficients_df <- coefficients_df[-1, ]
# Sort the features by the absolute value of the coefficients (feature importance)
coefficients_df$Abs_Coefficient <- abs(coefficients_df$Coefficient)
coefficients_df <- coefficients_df[order(-coefficients_df$Abs_Coefficient), ]
# Display top 10 most important features
head(coefficients_df, 10)
# Plot feature importance
library(ggplot2)
ggplot(coefficients_df, aes(x = reorder(Feature, Abs_Coefficient),
                            y = Abs_Coefficient)) +
  geom_bar(stat = "identity", fill = "skyblue") +
  coord_flip() +
  labs(title = "Feature Importance (Elastic Net)",
       x = "Feature",
       y = "Absolute Coefficient Value") +
  theme_minimal()
```

FOR ADDITIONAL CODE PLEASE VISIT: https://github.com/keshanagn/ANALYSIS-II